

Daniel Kwan

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor's Degree, Computer Engineering, Honours, Co-op

Sep 2025 – Present

- Clubs & Involvement: Computer Science Club, Data Science Club, Poker Club, Volleyball Intramurals GPA: 3.7/4.0

EXPERIENCE

Software Engineering Intern

May 2026 – Aug 2026

Aritzia

Vancouver, BC

- Incoming Summer 2026
- Business Technology Team

Software Developer

Jan 2026 – Present

Waterloo Aerial Robotics Group (Autonomy Subteam)

Waterloo, ON

- Engineered a fault-tolerant Python ground station using **MAVLink**, implementing a **multiprocessing architecture** with 4 concurrent worker processes for telemetry ingestion and command dispatch
- Optimized vision pipeline throughput by implementing **asynchronous frame processing** with **OpenCV**, maintaining a stable **30+** FPS for real-time red/blue target segmentation
- Reduced system heading error to **<5** degrees by designing a feedback control loop that computes relative yaw adjustments from live telemetry data

PROJECTS

Aegis (Yale Hack 2026 Finalist Solo) [🔗](#) | *FastAPI, scikit-learn, pytest, Typescript, Python*

- Delivered a scalable OSINT privacy platform that quantified identity exposure risks on a 0-100 scale by architecting secure REST APIs with **FastAPI** and **TypeScript**
- Achieved high-accuracy extraction of **6** distinct types of sensitive user data by developing a classification pipeline combining OCR, EXIF metadata parsing, and NER with TF-IDF cosine similarity scoring
- Enabled real-time monitoring of breach probability trends across **50+** concurrent user sessions by designing an interactive analytics dashboard backed by persistent cloud JSON storage

ExoDiscover (NASA Space Apps 2025) [🔗](#) | *Typescript, React, PyTorch, XGBoost, Lightcurve, Python*

- Secured a **Top 2%** (4th/250+) global finish at the NASA Space Apps Hackathon by leading a 5-member team to architect and deploy a full-stack exoplanet classification system
- Streamlined data ingestion for downstream model consumption—structuring **16** distinct transit-based feature sets—by engineering an automated time-series telemetry pipeline for Kepler and TESS datasets
- Delivered real-time machine learning inference results to end users by developing a RESTful **Flask** API integrated seamlessly with a responsive React frontend

Midas (QHacks 2026) [🔗](#) | *FastAPI, YOLOv11, OpenCV, Whisper, OpenAI*

- Ensured continuous, stutter-free frame rates on mobile AR clients by optimizing API request-response cycles for a **YOLOv11**-backed object detection backend
- Significantly increased model detection confidence in low-light environments by implementing a custom **OpenCV** preprocessing pipeline utilizing CLAHE contrast enhancement and sharpening filters
- Delivered a fully touchless, step-by-step repair guidance experience by engineering an asynchronous voice-interaction layer utilizing **FastAPI, Whisper STT, and OpenAI TTS**

SignCLI (UOttHack 2026) [🔗](#) | *PyTorch, Typescript, SQL, MediaPipe, Python*

- Built a real-time ASL translation engine capable of classifying 26 distinct gestures at live frame rates by engineering a low-latency **MediaPipe** vision pipeline trained on **70,000+** samples
- Established a reliable state management system that tracks historical per-letter accuracy across **100%** of user sessions by designing a structured **SQL** relational database schema
- Accelerated user skill acquisition by auto-generating custom practice sets targeted at lowest-performing signs through an adaptive algorithm that dynamically queries database records

TECHNICAL SKILLS

Languages: Python, C, C++, Typescript, JavaScript, Java, SQL (PostgreSQL)

Backend & Frameworks: FastAPI, Flask, React, Docker, Linux (RHEL), Git, MAVLink

Machine Learning & Data: PyTorch, OpenCV, scikit-learn, XGBoost, NumPy, Pandas, Jupyter